BookHaven Software Requirements Specification

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Revision History

Version	Date	Change	Status
1.0	2026-09-13	Original SRS, functional requirements, and user stories.	Baseline
1.1	2026-11-08	Added cart, checkout, order history, refunds, fulfillment, and admin oversight.	Superseded
1.2	2026-11-08	Defined the local development MVP and deferred production integrations.	Superseded
1.3	2026-11-11	Verified checkout and order behavior and documented limitations.	Superseded
1.4	2026-11-11	Added comparison, quantities, state tax, role homepages, seller earnings, digital-delivery retrieval, and admin order intervention.	Current

1 Introduction

1.1 Purpose

BookHaven is a web-based marketplace connecting independent booksellers with buyers. Sellers create listings and manage inventory. Buyers search, compare, purchase, review, and request refunds for books. Administrators moderate listings, oversee orders and refunds, and may intervene in marketplace order status.

1.2 Current Scope

The current release is a local PostgreSQL, Express, React, and Vite MVP. It supports development payment-format validation but does not connect to a real payment provider. The current implementation status is authoritative over earlier planned behavior where the two differ.

1.3 Definitions

Buyer
: A registered user who searches for and purchases listings.

Seller
: A registered user who creates listings and fulfills orders.

Administrator
: A platform user with moderation, refund, and order intervention privileges.

Listing
: A book offered by a seller, including description, price, condition, stock, and optional digital metadata.

Digital delivery
: A buyer-authorized access token and optional URL created for a digital listing. Production file hosting is outside the current MVP.

Tax state
: The two-letter US state used to select the configured base state tax rate at checkout.

2 System Features

2.1 Accounts and Access

- Buyers and sellers can register, log in, log out, and edit permitted profile information.
- Server-side role checks protect buyer, seller, and administrator routes.
- The current development session token has a fixed lifetime; inactivity does not reset its expiration timer.

2.2 Listings, Search, and Comparison

- Sellers can create, edit, and remove their own listings.
- ISBN-10 and ISBN-13 checksums are validated.
- Administrators approve or reject listings and provide rejection reasons.
- Only approved listings appear in buyer search and browse results.
- Buyers can search by title, author, genre, or ISBN.
- Buyers can compare two to four approved listings. The comparison includes title, author, ISBN, genre, description, condition, price, stock, seller, listing type, and image metadata.
- The comparison API is `GET /api/listings/compare?ids=<comma-separated-ids>`.

2.3 Cart, Quantities, Checkout, and Tax

- Buyers can add approved, in-stock listings to a cart, remove items, and edit whole-number quantities.
- Cart and buy-now checkout reject quantities below one or above available stock.
- Checkout validates the shipping address and development card format. Raw card data is not stored.
- Stock is locked and decremented transactionally when an order is created.
- Buyers select a two-letter US tax state. The configured base state rate is applied to the subtotal, rounded to cents, and saved with the order.
- Orders persist `tax_state`, `tax_rate`, `tax_amount`, `total_amount`, and the request IP address for audit purposes.
- An optional trusted mapping may be configured as `IP_STATE_MAP=address:STATE,address:STATE`. A matching trusted mapping overrides the selected state.

2.4 Refunds

- A buyer may submit a refund request for an eligible order and must provide a reason.
- Each seller associated with the order receives an in-app notification.
- Sellers and administrators may approve or reject refund requests.
- Buyers receive an in-app notification after the decision.
- The local order and refund status is updated. No external payment refund is issued by the current MVP.

2.5 Orders and Digital Delivery

- Buyers can view their own order history and order statuses.
- Sellers can view their own order items, process shipments, enter carrier and tracking information, and complete or cancel eligible orders.
- When a digital listing is purchased, checkout creates a `digital_deliveries` record with a unique access token.
- A buyer can retrieve delivery information only for an order they own through `GET /api/orders/:orderId/`
- The current MVP does not host protected digital files or provide a production download/content service.

2.6 Role Homepages and Seller Earnings

- `/home` routes an authenticated buyer, seller, or administrator to the corresponding role dashboard.
- Sellers can view gross sales, items sold, and order count in the seller dashboard and profile.
- Cancelled and refunded orders are excluded from seller gross sales.

2.7 Administrator Intervention

- Administrators can review listings, oversee refunds, and view platform orders.
- Administrators can override an order status to paid, processing, shipped, delivered, cancelled, or refunded.
- Each override creates an `admin_actions` audit record and notifies the buyer.

3 Requirements Status

3.1 Completed Requirements

The following current MVP capabilities are implemented: registration and login; role authorization; seller listing ownership; ISBN validation; listing moderation; search; listing details; cart operations; quantity validation; transactional checkout; state tax calculation and persistence; order history; refund requests and decisions; seller notifications; fulfillment; digital token creation and buyer-authorized retrieval; reviews; wishlists; comparison; role homepages; seller earnings; and administrator order intervention.

3.2 Partial Requirements

ID	Requirement	Current limitation
REQ-006	Profile maintenance	Password changes and the seller store-name workflow are incomplete.
REQ-007	Thirty-minute inactivity expiration	The development token has a fixed lifetime; activity does not reset expiration.
REQ-010	Re-review edited listings	Full automatic re-review behavior for every materially edited field is incomplete.
REQ-024	Grouped genre browsing	Genre filtering exists, but a dedicated grouped browse interface is not implemented.
REQ-030	Itemized pre-payment review	Checkout does not yet provide a separate complete order-summary step before submission.
REQ-031	Payment processing	Only local card-format validation exists; no real payment gateway is connected.
REQ-033	Failed payment rollback	Database and stock rollback exist, but provider-decline simulation is not implemented.
REQ-035	Refund expiration	No configurable return/refund eligibility window is enforced.
REQ-043	Seller cancellation refund	Local refund state is supported, but no real provider refund is issued.
REQ-045	Digital delivery	Token creation and buyer retrieval work; protected file hosting/download delivery is not implemented.
REQ-052	Review display	Individual reviews are available, but average-rating calculation is incomplete.

3.3 Deferred Requirements

- Real payment gateway integration and provider-based refunds.

- Automatic public-IP geolocation. The application stores IP audit data and supports trusted static mappings, but does not send IP addresses to an external geolocation service.
- County, city, district, exemption, and tax-jurisdiction calculations. The current tax feature uses configured base state rates and buyer-selected state, with trusted static IP mapping as an optional override.
- Production digital-file hosting, protected downloads, and content delivery.
- Production hosting, email provider, HTTPS deployment, and mobile applications.
- Full browser E2E and Newman assertion coverage.

4 User Stories

4.1 Purpose and Interpretation

This chapter records the stakeholder-facing user stories that define the BookHaven MVP. Each story identifies the actor's goal, preconditions, normal flow, alternate flows, and acceptance criteria. A story may describe behavior that is planned but not fully implemented; its implementation status is defined in Chapter 4. In particular, payment-provider processing, public-IP geolocation, return-expiration enforcement, and production digital-file delivery are explicitly partial or deferred rather than silently assumed to be complete.

4.2 Buyer User Stories

4.2.1 User Story 1 — Register and log in

As a visitor, I want to create a Buyer or Seller account and log in so that I can use the marketplace features assigned to my role.

Preconditions:

- The visitor is not already authenticated.
- The submitted email address is not already registered.

Main flow:

1. The visitor opens the registration page.
2. The visitor enters a name, email address, password, and account role.
3. The system validates required fields and password format.
4. The system rejects duplicate email addresses.
5. The system hashes the password and creates the account.
6. The visitor logs in with the new credentials.
7. The system issues a development session token and redirects by role.

Alternate flows:

- If a required field is missing, the system identifies the field and does not create the account.
- If the email already exists, the system returns a generic registration error and does not overwrite the existing account.
- If the password is incorrect at login, the system returns a generic invalid-login message without revealing whether the email or password failed.

Acceptance criteria:

- A valid Buyer and a valid Seller can register independently.

- Passwords are never stored in plaintext.
- Invalid credentials do not expose sensitive account information.
- A user cannot select or obtain administrator privileges during registration.

4.2.2 User Story 2 — Maintain a profile

As an authenticated user, I want to edit permitted account information so that my profile remains accurate.

Preconditions: The user is logged in and owns the profile being edited.
Main flow:

1. The user opens Profile.
2. The system displays editable account fields.
3. The user changes permitted information such as name or email.
4. The system validates the values and checks email uniqueness.
5. The system saves the changes and displays confirmation.

Alternate flows:

- An email already assigned to another account is rejected.
- Account type is not displayed as an editable field.
- Password-change and store-name workflows are retained as future profile enhancements where they are not available in the current MVP.

4.2.3 User Story 3 — Use the role homepage

As an authenticated user, I want a homepage that returns me to the features for my role so that I can quickly resume marketplace work.

Main flow:

1. The user opens `/home`.
2. The system reads the authenticated role.
3. Buyers are routed to the Buyer dashboard, Sellers to the Seller dashboard, and Administrators to the Administrator dashboard.
4. The dashboard provides navigation to the role's available features.

Alternate flows:

- An unauthenticated visitor is routed to login.
- A user attempts to open a dashboard for another role; server-side authorization rejects the request.

4.2.4 User Story 4 — Search and browse books

As a buyer, I want to search and browse approved books by title, author, genre, or ISBN so that I can find items to purchase.

Preconditions: Approved listings exist or the system can display a no-results state.

Main flow:

1. The buyer enters a search term or selects a genre.
2. The system searches only approved listings.
3. The system sorts matching results alphabetically by title.
4. The buyer views title, author, genre, condition, price, stock, seller, and image information.
5. The buyer opens a listing for its complete description and details.

Alternate flows:

- A search with no matches displays a clear "No results found" message.
- Unapproved, rejected, or removed listings are excluded.
- A dedicated grouped genre-browse interface remains a partial enhancement; genre filtering is available.

4.2.5 User Story 5 — Compare marketplace listings

As a buyer, I want to compare two to four approved listings so that I can choose the best item from different sellers or conditions.

Preconditions:

- At least two approved listings are available.
- The buyer has selected no more than four distinct listing identifiers.

Main flow:

1. The buyer selects comparison checkboxes in the catalog.
2. The buyer chooses Compare.
3. The system validates that two to four valid listings were selected.
4. The system retrieves the approved listings.
5. The comparison page displays title, author, ISBN, description, genre, condition, price, stock, seller, image, and listing type in a consistent table.
6. The buyer returns to the catalog or opens a selected listing.

Alternate flows:

- Fewer than two valid items produces a validation message.
- More than four selections are limited to the supported comparison size.
- An unapproved or missing listing is not presented as an approved comparison result.

Acceptance criteria: The endpoint `GET /api/listings/compare?ids=<ids>` and the Buyer comparison page use the same approved-listing rules and expose the description as well as the commercial comparison fields.

4.2.6 User Story 6 — Add and remove cart items

As a buyer, I want to add and remove books from my cart so that I can adjust my intended purchase before checkout.

Preconditions: The buyer is authenticated and the listing is approved and in stock.

Main flow:

1. The buyer selects Add to Cart on an approved listing.
2. The system creates or updates the buyer's cart item.
3. The cart displays the item, unit price, quantity, and subtotal.
4. The buyer removes an item when it is no longer wanted.
5. The system recalculates the subtotal and cart contents.

Alternate flows:

- An unauthenticated buyer is asked to log in.
- A zero-stock listing cannot be added.
- Removing the last item displays an empty-cart message and prevents checkout until another item is added.

4.2.7 User Story 7 — Select item quantities

As a buyer, I want to select how many copies of an item I want so that I can purchase multiple available copies without exceeding stock.

Main flow:

1. The buyer enters a whole-number quantity in the cart or Buy Now form.
2. The system validates that the quantity is at least one.
3. The system compares the requested quantity with current stock.
4. The system updates the cart quantity or proceeds to checkout.
5. The system multiplies unit price by quantity when calculating subtotal.

Alternate flows:

- Decimal, zero, negative, or non-numeric quantities are rejected.
- A quantity greater than stock returns an error and leaves the prior quantity unchanged.
- If stock changes before checkout, the transactional checkout rejects the order rather than creating an oversold order.

4.2.8 User Story 8 — Review an order before checkout

As a buyer, I want to review my selected items, quantities, prices, shipping address, tax, and total before submitting checkout so that I can correct mistakes.

Main flow:

1. The authenticated buyer opens checkout with a non-empty cart.
2. The system displays the cart items and quantities.

3. The buyer enters a shipping address and selects a tax state.
4. The system displays the subtotal, configured tax rate, tax amount, and calculated total.
5. The buyer enters development card-format information.
6. The buyer submits the order.

Status note: Cart and checkout values are implemented, but a separate fully itemized pre-payment review screen remains a partial UI enhancement.

4.2.9 User Story 9 — Calculate state tax

As a buyer, I want the checkout tax to reflect my selected state so that the amount recorded for the order is understandable.

Main flow:

1. The buyer selects a valid two-letter US state.
2. The system selects the configured base rate for that state.
3. The system calculates tax from the subtotal and rounds the amount to cents.
4. The system displays the tax and includes it in the final total.
5. The system saves tax state, rate, amount, total, and request IP audit data with the order.

Location and privacy rules:

- A trusted deployment may define `IP_STATE_MAP` entries for known addresses; a matching entry overrides the selected state.
- The system does not claim that an IP address proves a buyer's location.
- Automatic public-IP geolocation and local tax-jurisdiction lookup are deferred. City, county, district, exemption, and tax-holiday rules are not included in the base MVP calculation.

4.2.10 User Story 10 — Complete checkout

As a buyer, I want a successful checkout to create my order and reserve stock so that my purchase is recorded.

Main flow:

1. The buyer submits valid checkout information.
2. The system begins a database transaction and locks the relevant stock updates.
3. The system verifies that every item is approved and available.
4. The system decrements stock and creates the order and order items.
5. The system creates digital-delivery records where applicable.
6. The system clears the cart after commit and shows confirmation.
7. The system sends an in-app new-order notification to the seller.

Alternate flows:

- Invalid card format, address, state, quantity, or product data is rejected before order creation.
- A stock conflict rolls back the transaction and leaves the order uncreated.
- A database failure rolls back stock and order changes.
- A real payment-provider decline cannot currently be simulated because a provider is deferred.

4.2.11 User Story 11 — View order history

As a buyer, I want to view my own orders and statuses so that I can track purchases and use eligible post-purchase actions.

Main flow:

1. The buyer opens Order History.
2. The system authenticates the buyer.
3. The system queries only orders owned by that buyer.
4. The system displays order status, total, date, items, quantities, unit prices, and fulfillment status.

Alternate flows:

- A missing or invalid token returns Unauthorized.
- A buyer with no orders receives an empty history rather than another buyer's order data.

4.2.12 User Story 12 — Request a refund

As a buyer, I want to request a refund with a reason so that a seller or administrator can review a problem with my order.

Preconditions: The buyer owns the order and the order is in an eligible local status.

Main flow:

1. The buyer selects Request Refund from Order History.
2. The buyer enters a reason.
3. The system validates the order identifier, buyer ownership, eligibility, and non-empty reason.
4. The system creates a refund request and marks the local order refund status as requested.
5. The system notifies every seller associated with the order.
6. The buyer receives a request confirmation.

Alternate flows:

- Missing reason returns a validation error.
- A nonexistent order returns Not Found.
- An order belonging to another buyer cannot be used.
- A duplicate or ineligible request is rejected.
- A configurable time-based return window is a deferred enhancement.

4.2.13 User Story 13 — Receive a refund decision

As a buyer, I want to know whether my refund request was approved or rejected so that I understand the final marketplace decision.

Main flow:

1. A seller or administrator reviews the request.
2. The reviewer approves or rejects the request.
3. The system updates the refund request and local order status.
4. The system records the reviewer and decision time where applicable.
5. The system sends an in-app notification to the buyer.
6. The buyer sees the decision in Order History.

Status note: The current implementation updates local state only; it does not send money through a payment provider.

4.2.14 User Story 14 — Access a digital order

As a buyer, I want to retrieve the access token or URL for a digital order that I own so that I can access its delivery.

Main flow:

1. The buyer completes checkout for an approved digital listing.
2. The system creates a unique delivery token tied to the order, product, and buyer.
3. The buyer opens the digital-delivery action in Order History.
4. The system verifies buyer ownership.
5. The system returns the token and configured URL when available.

Alternate flows:

- A buyer cannot retrieve another buyer's delivery.
- A missing delivery returns an error rather than an invented token.
- Production file hosting, protected download streaming, licensing, and download limits remain deferred.

4.2.15 User Story 15 — Write a review

As a buyer, I want to rate and review a delivered book so that I can share my experience with other buyers.

Preconditions: The buyer owns a delivered order containing the book and has not already reviewed it.

Main flow:

1. The buyer opens the delivered order or product review form.
2. The buyer selects a rating from one through five stars.
3. The buyer enters review text.

4. The system checks authentication, purchase, delivery, duplicate status, rating range, and prohibited content.
5. The system stores the valid review.
6. The product page displays the review.

Alternate flows:

- A buyer without a delivered purchase is rejected.
- A duplicate review is rejected.
- A missing or out-of-range rating is rejected.
- Prohibited content is rejected with an explanation.
- Average-rating calculation remains a partial enhancement.

4.2.16 User Story 16 — Manage a wishlist

As a buyer, I want to save and remove books from a wishlist so that I can find items later without placing them in my cart.

Main flow:

1. The buyer selects Add to Wishlist for a listing.
2. The system saves the buyer-listing relationship.
3. The buyer opens the wishlist and sees all saved items.
4. The buyer removes an item when desired.
5. The system confirms the updated wishlist.

Alternate flows:

- Duplicate additions are rejected with feedback.
- Unavailable listings remain visible and are marked unavailable.
- The buyer can remove an unavailable listing.

4.3 Seller User Stories

4.3.1 User Story 17 — Create a physical listing

As a seller, I want to list a physical book with complete information so that buyers can discover and purchase it.

Preconditions: The seller is authenticated and owns the information and image being submitted.

Main flow:

1. The seller opens My Listings and selects Add New Listing.
2. The seller chooses Physical and enters title, author, ISBN, genre, condition, description, price, and stock.
3. The seller optionally uploads a supported cover image.

4. The system validates required fields, ISBN checksum, price, and stock.
5. The system creates the listing in pending review.
6. An administrator receives the listing in the moderation queue.

Alternate flows:

- Missing fields, invalid ISBN, invalid price, or invalid stock prevent submission.
- An unsupported or oversized image is rejected.
- A seller cannot create a listing on behalf of another seller.

4.3.2 User Story 18 — Create a digital listing

As a seller, I want to identify a listing as digital so that buyers can receive digital-order access after purchase.

Main flow:

1. The seller selects Digital listing type.
2. The system records the listing type and digital URL when configured.
3. Physical-only fields are omitted or marked not applicable.
4. The seller submits the listing for administrator review.
5. The administrator approves or rejects the listing.

Status note: The current MVP supports digital listing metadata, checkout token creation, and buyer-authorized retrieval. It does not implement production manuscript hosting, DRM, protected downloads, or usage limits.

4.3.3 User Story 19 — Edit and remove a listing

As a seller, I want to edit or remove my own listing so that inventory and product information remain current.

Main flow:

1. The seller opens one of their listings.
2. The seller changes permitted fields or chooses Remove.
3. The system verifies seller ownership.
4. The system validates changes and persists them.
5. The system hides a removed listing from buyers.

Alternate flows:

- A seller cannot edit or remove another seller's listing.
- A materially changed listing should return to moderation; complete automatic re-review coverage remains partial.

4.3.4 User Story 20 — View seller orders

As a seller, I want to view orders containing my listings so that I can fulfill only the items I sold.

Main flow:

1. The seller opens Pending Orders.
2. The system returns order items associated with that seller.
3. The seller sees order identifier, title, quantity, fulfillment status, buyer shipping information permitted for fulfillment, and order information.
4. The seller selects an order item for processing.

Alternate flows:

- A seller cannot view unrelated sellers' order items.
- An unauthenticated or non-seller request is rejected.

4.3.5 User Story 21 — Fulfill and ship a physical order

As a seller, I want to process and ship a physical order with tracking so that the buyer can follow delivery.

Main flow:

1. The seller marks an eligible item as Processing.
2. The seller enters a carrier and tracking number.
3. The system changes the fulfillment status to Shipped.
4. The system records the tracking data.
5. The system notifies the buyer.
6. The seller later marks the item Delivered when appropriate.

Alternate flows:

- Invalid transitions are rejected.
- Missing tracking data is rejected when shipping requires it.
- A seller cannot fulfill an item not associated with their account.

4.3.6 User Story 22 — Cancel an unshipped order

As a seller, I want to cancel an order I cannot fulfill so that the buyer is informed and the local order state is corrected.

Main flow:

1. The seller selects Cancel on an eligible unshipped order.
2. The seller enters a cancellation reason.
3. The system changes the local status to cancelled.
4. The system records the local refund state where applicable.
5. The buyer receives the cancellation reason.

Status note: A real payment refund to the original payment method is deferred because no payment provider is connected.

4.3.7 User Story 23 — Review seller earnings

As a seller, I want to see gross sales, items sold, order count, and order information in my dashboard and profile so that I can track earnings.

Main flow:

1. The seller opens the Seller dashboard or Profile.
2. The system requests the seller's earnings summary.
3. The system aggregates eligible order items for that seller.
4. The system displays gross sales, items sold, and order count.
5. The seller reviews order prices and quantities through the seller order list.

Acceptance criteria:

- Earnings are scoped to the authenticated seller.
- Cancelled and refunded orders are excluded from gross sales.
- A seller cannot request another seller's earnings.

4.3.8 User Story 24 — Receive refund-request notices

As a seller, I want to receive a notification when a buyer requests a refund for one of my items so that I can review the request.

Main flow:

1. A buyer submits a valid refund request.
2. The system identifies every seller represented in the order.
3. The system creates an in-app notification for each relevant seller.
4. The seller opens the refund queue and reviews the reason.
5. The seller approves or rejects the request.

4.4 Administrator User Stories

4.4.1 User Story 25 — Moderate listings

As an administrator, I want to approve or reject pending listings so that only appropriate and accurate listings become visible.

Main flow:

1. The administrator opens the pending listing queue.
2. The system orders pending entries by submission date.
3. The administrator reviews listing fields, ISBN, and prohibited-content flags.
4. The administrator approves a valid listing or rejects it with a reason.
5. The system changes visibility and notifies the seller.

Alternate flows:

- An invalid ISBN prevents approval.
- Prohibited content causes a moderation warning or rejection.
- A non-administrator cannot access moderation actions.

4.4.2 User Story 26 — Manage prohibited words

As an administrator, I want to maintain prohibited words and phrases so that listings and reviews can be screened consistently.

Main flow:

1. The administrator opens prohibited-content management.
2. The administrator adds, edits, or removes a word or phrase.
3. The system validates and saves the moderation rule.
4. Future listing and review submissions use the maintained rule set.

4.4.3 User Story 27 — Oversee refunds

As an administrator, I want to review all refund requests and decide disputed requests so that I can intervene between buyers and sellers.

Main flow:

1. The administrator opens the platform refund queue.
2. The system displays buyer, order, reason, seller, and refund status.
3. The administrator approves or rejects the request.
4. The system updates local refund and order status.
5. The system notifies the buyer and records the administrative action.

4.4.4 User Story 28 — Override order status

As an administrator, I want to override an eligible order status so that I can resolve operational problems between a buyer and seller.

Main flow:

1. The administrator opens platform order oversight.
2. The administrator selects an order and an allowed status: `paid`, `processing`, `shipped`, `delivered`, `cancelled`, or `refunded`.
3. The system validates the administrator role and order identifier.
4. The system updates the order.
5. The system creates an `admin_actions` audit record.
6. The system sends an in-app notification to the buyer.

Alternate flows:

- An invalid status or order identifier is rejected.
- A buyer, seller, or anonymous caller cannot use the administrator endpoint.
- The administrator cannot claim that a real payment was refunded when the payment-provider integration is deferred; the local status must be understood as a marketplace record.

4.4.5 User Story 29 — Review platform earnings

As an administrator, I want to review platform-wide order and earning aggregates so that I can monitor marketplace activity.

Main flow:

1. The administrator opens the administrative dashboard.
2. The system aggregates eligible order totals and counts.
3. The system excludes cancelled and refunded orders from gross-sales aggregates where configured.
4. The administrator reviews the summary without seeing raw payment-card data.

4.5 Cross-Cutting Acceptance Criteria

- Every protected story requires authentication and server-side role authorization; hiding a frontend link alone is insufficient.
- Buyer data, orders, digital deliveries, cart items, and refunds are scoped to the authenticated owner or authorized role.
- Listing and order state changes return explicit errors for invalid input instead of silently succeeding.
- Stock updates used by checkout are transactional and cannot create an order when the requested quantity is unavailable.
- Tax totals are calculated from the persisted subtotal and configured state rate; the request IP is audit metadata, not a guaranteed location source.
- Refund and administrator actions notify affected users through the existing in-app notification mechanism.
- Digital delivery access is authorized by buyer ownership and is not treated as production file hosting.
- The requirements traceability matrix in `docs/RTM.csv` maps the original requirements and added requirements REQ-058 through REQ-063 to their implementation and test cases.

5 Verification

- Backend TypeScript build: passed.
- Frontend production build: passed.
- Frontend lint: passed.
- Backend tests: 23 passed, 0 failed against the isolated PostgreSQL test database.
- New feature test cases TC-024 through TC-027 are documented in `docs/TestCases.md`; dedicated seeded integration fixtures remain follow-up coverage.

6 Traceability Notes

The detailed requirement traceability matrix is maintained in `docs/RTM.csv`. Requirements REQ-058 through REQ-063 cover comparison, quantity controls, state tax, role homepages, seller earnings, and administrator order intervention. Digital delivery is tracked as partial because token creation and authorized retrieval are implemented while production content delivery is deferred.